6(a)	<u>sum of</u> current(s) entering a junction = <u>sum of</u> current(s) leaving (the same junction) **or** (algebraic) sum of current (s) at a junction is zero	**B1**
6(b)	(by Kirchhoff's second law) $V = V_1 + V_2$	**B1**
	so $IR_T = IR_1 + IR_2$ (and cancelling I gives) $R_T = R_1 + R_2$ **or** $V / I = V_1 / I + V_2 / I$ (and substituting R gives) $R_T = R_1 + R_2$	**B1**
6(c)	current in circuit = 1.35 / (10 + 15) (= 0.054 A)	**C1**
	$r = (E - V) / I$	**C1**
	= (1.5 – 1.35) / 0.054 = 2.8 Ω	**A1**
	or	
	by potential divider principle $\dfrac{0.15}{1.35} = \dfrac{r}{25}$	**(C2)**
	r = 2.8 Ω	**(A1)**
	or	
	I = 1.35 / (10 + 15) (= 0.054 A)	**(C1)**
	total resistance = 1.50 / 0.054 (= 27.8 Ω) r = 27.8 – 25	**(C1)**
	r = 2.8 Ω	**(A1)**

Question	Answer	Marks
6(d)(i)	the (total) resistance (of the circuit) has decreased (and e.m.f. is unchanged)	**M1**
	(the current (in the cell) will) increase	**A1**
6(d)(ii)	(as the current is greater and so there is a) larger p.d. across the internal resistance	**M1**
	(terminal p.d. will) decrease	**A1**